

OV519 CAMERAMATE™ Processor

Enhanced Single Chip Processor for USB PC Camera Applications

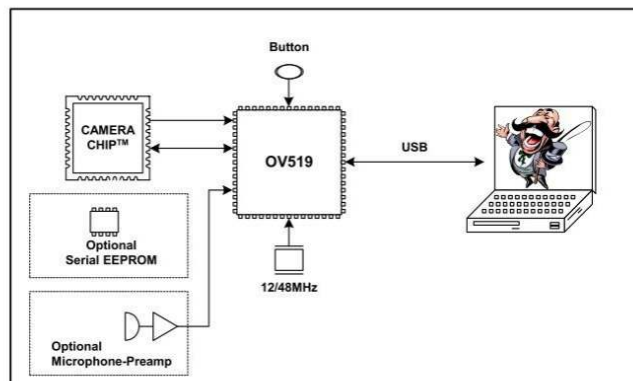
General Description

The OV519 CAMERAMATE™ Processor is an enhanced single chip processor for high-resolution USB PC camera applications. When combined with an OmniVision CAMERACHIP™ color image sensor, such as the OV86xx (SVGA), OV76xx (VGA) or the OV66xx (CIF), the OV519 comprises a low cost, integrated USB camera system, with no additional USB transceiver or DRAM required. As shown in the system block diagram below, the OV519 also supports an audio input with for full audio/video operation.

During operation, the OV519's camera interface synchronizes with the CAMERACHIP YC_bC_r 422 progressive video data to perform Down-sampling, Clamping and Windowing (DCW) functions with the desired resolution requested by users through the USB command set. The embedded Data Compression Engine then compresses the video data to standard JPEG format and the resultant video data is transferred to the host through USB.

This standard USB interface allows the end user to implement PC camera systems easily, rapidly reducing the time-to-market.

PC Camera System Block Diagram

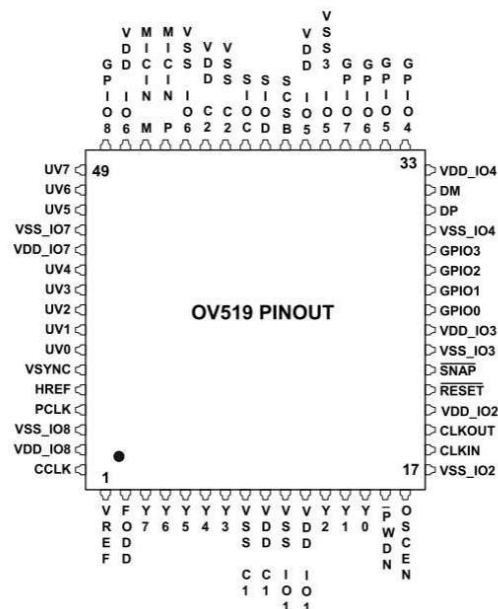


Ordering Information

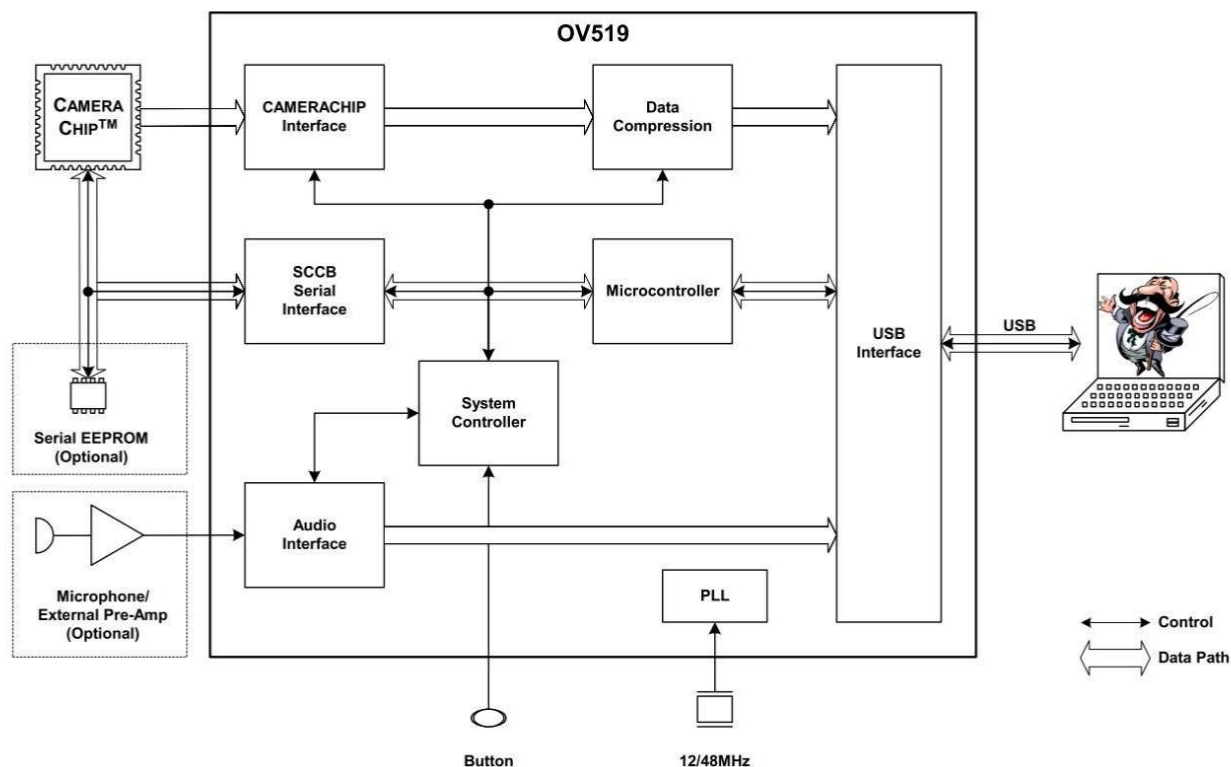
Product	Package	Temperature Range
OV519	TQFP-64	0° to 70°C

Features

- Low cost, single chip solution for high resolution USB PC camera applications.
- Full USB support (v1.1) including 2 selectable default hardwired video/audio descriptors as well as options for additional customer-specific USB PIDs.
- High-performance video transfer rate of 20FPS for SVGA and 30FPS for VGA, CIF/SIF & QCIF/QSIF formats with selectable YUV 4:2:2, 4:2:0, 4:1:1 or 4:0:0 formatting.
- Full USB audio interface (Class 1.0 compliance) incorporating a 16-Bit mono ADC for microphone usage with sampling rates of 8KHz, 16KHz and 24KHz.
- Low voltage 2.5V (Core) and 3.3V (I/O) operation.
- 8-Bit 8051-compatible microcontroller, with embedded 512-byte RAM
- External system clock or independent crystal operation.
- Down-sampling, Clamping and Windowing (DCW) functions
- YUV averaging and Y distribution with Y enhancement
- Defect compensation up to 8 pixels in YUV input



Functional Block Diagram



CAMERACHIP Families Supported

- OV86xx (SVGA)
- OV76xx (VGA)
- OV66xx (CIF)

Peripherals Supported

External EEPROM (Maximum size: 4K Bits):

An external serial EEPROM can be used to store information such as pixel defect addresses (maximum of 8 addresses for Y & UV, separately) and additional USB descriptors specified by the OEM or user-specific applications.

External Microphone and Preamp:

(Shown in functional block diagram)

Function Block Details

- **CAMERACHIP Interface:**

The camera data interface supports both 8-bit and 16-bit data widths at clock speeds up to 48MHz, as a function of the camera chip used.

- **Audio Interface:**

The audio input accepts the output of a microphone/pre-amp and converts it to a 16 Bit digital datastream with user-selected sampling rates of 8KHz, 16KHz or 24KHz. This digital audio data is then transferred to the PC system through the USB interface, providing full video & audio to the end user.

- **Microcontroller:**

The microcontroller operates as an embedded controller only and can be programmed by the user if custom applications are desired. ***Please contact OmniVision for further information on microcontroller implementation.***

- **System Controller:**

This function provides general system control to the internal logic. This includes all clock and reset signals and functions such as controlling the USB Power-Down SUSPEND mode and GPIO (General Purpose I/O) configuration, which can allow the user to control external peripherals.

- **Data Compression:**

The OV519 uses a standard JPEG compression engine to transmit high-resolution images at video rates.

- **SCCB Serial Interface:**

The SCCB interface controls the CAMERACHIP operation as well as interfacing to the optional EEPROM. In all cases, the OV519 is considered the SCCB 'Master' and all other SCCB devices, such as the CAMERACHIP or optional EEPROM, are designated as 'Slaves'.

Note: *The description below is a functional overview on the interface. For a complete description, please refer to Application Note 'AN101 – SCCB Functional Specification'.*

SCCB WRITE (To Slave) Transmission:

BYTE (Register) WRITE																										
7	6	5	4	3	2	1	0	X	7	6	5	4	3	2	1	0	X	7	6	5	4	3	2	1	0	X
<u>Phase 1</u>									<u>Phase 2</u>									<u>Phase 3</u>								
Slave Device Identifier [Master-to-Slave]									Slave Address (Register) [Master-to-Slave]									Data [Master-to-Slave]								

- SCCB Serial Interface (Continued):**

Phase 1: [Bits 7:1] Slave (CAMERACHIP or optional EEPROM) device identifier, which is factory-assigned to each device on the SCCB bus so the Master (OV519) can identify and communicate with all Slave devices. Because of this, each CAMERACHIP family member has its own identifier, such as the OV7620 which is defined as 0x42.

[Bit 0] Read/Write [R/W] Bit = 0 (WRITE)

Phase 2: Slave address (register)

Phase 3: WRITE data

Example: Slave ID-R/W: 0x42 (0100 0010) [OV7620, Bit 0 = WRITE]
 Slave Address: 0x22 (0010 0010) [FRAR Register]
 Data: 0x11 (0001 0001)
 SCCB WRITE: 0100 0010 [X] 0010 0010 [X] 0001 0001 [X]

SCCB READ (From Slave) Transmission (Slave-to-Master):

BYTE (Register) READ																																			
7	6	5	4	3	2	1	0	X	7	6	5	4	3	2	1	0	X	7	6	5	4	3	2	1	0	X	7	6	5	4	3	2	1	0	X
<u>Phase 1A</u>									<u>Phase 2</u>									<u>Phase 1B</u>									<u>Phase 3</u>								
Slave Device Identifier [Master-to-Slave]									Slave Address (Register) [Master-to-Slave]									Slave Device Identifier [Master-to-Slave]									Data [Slave-to-Master]								
Set Internal Slave Address																		Read Slave Data at Set Address																	

Phase 1A: [Bits 7:1] Slave (CAMERACHIP or optional EEPROM) Device Identifier
 [Bit 0] Read/Write [R/W] Bit = 0 (WRITE)

Phase 2: Slave address (internal register)

Phase 1B: [Bits 7:1] Slave (CAMERACHIP or optional EEPROM) Device Identifier
 [Bit 0] Read/Write [R/W] Bit = 1 (READ)

Phase 3: READ data

Example: Slave ID-R/W: 0x42 (0100 0010) [OV7620, Bit 0 = WRITE]
 Slave Address: 0x22 (0010 0010) [FRAR Register]
 Slave ID-R/W: 0x43 (0100 0011) [OV7620, Bit 0 = READ]
 Data: [In register]
 SCCB WRITE: 0100 0010 [X] 0010 0010 [X] 0100 0011 [X] [Slave Data] [X]

- Phase-Locked Loop (PLL):**

The internal PLL is used for CAMERAMATE synchronous clocking and may be enabled or disabled, as the user requires.

CAMERAMATE™ Clocking Options		
Option	GPIO[6]	Internal PLL
12MHz xtal or clock-signal operation	0	Enabled
48MHz xtal or clock-signal operation	1	Disabled

• **USB Interface:**

The OV519 uses two default hardwired descriptors for video or audio/video operation, and, additionally, has the option of another 16 user-selectable product IDs using the GPIO[3:0] pins. For OEM manufacturers, further USB flexibility can be obtained by incorporating a serial EEPROM to store additional descriptor information. ***If an additional PID is required, OmniVision will issue product IDs (PIDs) on a first-come, first-served basis.***

Hardwired USB Descriptors			
Descriptor	Function	GPIO[5:4]	Properties
1	Video only	00	One configuration, One Interface: Video interface only
0	Video and Audio	01 10 11	One configuration, Three Interface Options: - Interface 0 – Video - Interface 1 – Audio Control - Interface 2 – Audio Stream
Note		GPIO[5:4] default = 01 (See STROBES section for more details)	

• **Strobes:**

The OV519 uses 8 Strobe pins (GPIO[7:0]) to control the system (An additional pin GPIO[8] is present, but not used for the strobes).

Strobe Functionality				
Pin	Name	Pin Type	Function	
36	GPIO[7]	I	Microcontroller Enable	0: Disable [DEFAULT] 1: Enable
35	GPIO[6]	I	PLL Bypass	0: Internal PLL Enable [DEFAULT] 1: Internal PLL Disable
34-33	GPIO[5:4]	I	Audio Control	0: (00) Disable audio 1: (01) 16Kbps sampling [DEFAULT] 2: (10) 8Kbps sampling 3: (11) 24Kbps sampling
28-25	GPIO[3:0]	I	Strobe[7] = 0 Microcontroller Disabled	Support for 16 additional USB PIDs
			Strobe[7] = 1 Microcontroller Enabled	Strobe[3]: Emulator Enable Strobe[2]: Select Boot ROM Strobe[1]: Reserved Strobe[0]: Reserved

Note: All GPIO pins are read by the OV519 at Power-On. Therefore, after VDD supplies have settled, further strobe pin changes will not be seen by the OV519, so if configuration changes are required, the user must cycle the system power supplies to make the change.

In addition, when the GPIO pins are programmed as inputs, the user can select either an edge-triggered mode, with programmable positive/negative edge triggering, or a level-triggered mode using the System Controller registers.

Pin Description

Pin Description			
Pin	Name	Pin Type	Function
SYSTEM CONTROL			
15	PWDN	O	Power-Down (Enable USB Power-Down SUSPEND mode)
16	OSCEN	O	Oscillator Enable
18	CLKIN	I	Crystal/Oscillator clock input
19	CLKOUT	O	Crystal/Oscillator clock output
21	RESET	I	Power-On Reset
22	SNAP	I	Snapshot button input
SENSOR INTERFACE			
1	VREF	I/O	Vertical reference input from camera
2	FODD	I/O	Even/Odd field indication input from camera
39	SCSB	O	Serial camera control bus chip select output
40	SIOD	I/O	Serial camera control bus data
41	SIOC	O	Serial camera control bus clock output
59	VSYNC	I/O	Vertical sync input from camera
60	HREF	I/O	Horizontal reference input from camera
61	PCLK	I/O	Pixel clock input from camera
64	CCLK	O	Camera clock output
CAMERACHIP Y DATA			
3	Y[7]	I/O	Y Bit [7]
4	Y[6]	I/O	Y Bit [6]
5	Y[5]	I/O	Y Bit [5]
6	Y[4]	I/O	Y Bit [4]
7	Y[3]	I/O	Y Bit [3]
12	Y[2]	I/O	Y Bit [2]
13	Y[1]	I/O	Y Bit [1]
14	Y[0]	I/O	Y Bit [0]
CAMERACHIP UV DATA			
49	UV[7]	I/O	UV Bit [7]
50	UV[6]	I/O	UV Bit [6]
51	UV[5]	I/O	UV Bit [5]
54	UV[4]	I/O	UV Bit [4]
55	UV[3]	I/O	UV Bit [3]
56	UV[2]	I/O	UV Bit [2]
57	UV[1]	I/O	UV Bit [1]
58	UV[0]	I/O	UV Bit [0]
USB INTERFACE			
30	DP	I/O	USB D+
31	DM	I/O	USB D-

Pin Description (Continued)

Pin Description (Continued)			
Pin	Name	Pin Type	Function
AUDIO INTERFACE			
45	MICIN_P	I	Microphone Input (+)
46	MICIN_M	I	Microphone Input (-)
GENERAL PURPOSE I/O			
25	GPIO[0]	I/O	General Purpose I/O Bit [0]
26	GPIO[1]	I/O	General Purpose I/O Bit [1]
27	GPIO[2]	I/O	General Purpose I/O Bit [2]
28	GPIO[3]	I/O	General Purpose I/O Bit [3]
33	GPIO[4]	I/O	General Purpose I/O Bit [4]
34	GPIO[5]	I/O	General Purpose I/O Bit [5]
35	GPIO[6]	I/O	General Purpose I/O Bit [6]
36	GPIO[7]	I/O	General Purpose I/O Bit [7]
48	GPIO[8]	I/O	General Purpose I/O Bit [8]
POWER AND GROUND			
8	VSS_C1	GND	Core GROUND
9	VDD_C1	VDD	Core VDD
42	VSS_C2	GND	Core GROUND
43	VDD_C2	VDD	Core VDD
10	VSS_IO1	GND	I/O GROUND
11	VDD_IO1	VDD	I/O VDD
17	VSS_IO2	GND	I/O GROUND
20	VDD_IO2	VDD	I/O VDD
23	VSS_IO3	GND	I/O GROUND
24	VDD_IO3	VDD	I/O VDD
29	VSS_IO4	GND	I/O GROUND
32	VDD_IO4	VDD	I/O VDD
37	VSS_IO5	GND	I/O GROUND
38	VDD_IO5	VDD	I/O VDD
44	VSS_IO6	GND	I/O GROUND
47	VDD_IO6	VDD	I/O VDD
52	VSS_IO7	GND	I/O GROUND
53	VDD_IO7	VDD	I/O VDD
62	VSS_IO8	GND	I/O GROUND
63	VDD_IO8	VDD	I/O VDD

Electrical Characteristics

DC Electrical Characteristics						
Symbol	Parameter	Condition	Min	Typical	Max	Unit
VDD_IO	DC supply voltage – I/O	3.3V $\pm$ 10%	3.0	3.3	3.6	V
VDD_L	DC supply voltage – Core	2.5V $\pm$ 10%	2.25	2.5	2.75	V
V _{IH}	High level input voltage	CMOS	0.7 x VDD_IO			V
V _{IL}	Low level input voltage	CMOS			0.3 x VDD_IO	V
V _{OH}	High level output voltage	CMOS	0.9 x VDD_IO			V
V _{OL}	Low level output voltage	CMOS			0.1 x VDD_IO	V
I _{DDs}	SUSPEND Mode Current			50		μ A
I _{DDA}	Active (Operating) Current			40		mA
I _L	Input/Output Leakage	GND $\rightarrow$ VDD_IO			10	μ A

Absolute Maximum Ratings	
Ambient Storage Temperature	-40°C $\rightarrow$ +125°C
All Input/Output Voltages with Respect to Ground	-0.3V $\rightarrow$ VDD_IO+1V
Lead Temperature (Soldering, 10 seconds)	+300°C
ESD Rating	2000V (maximum)

Device Control Registers

Note ENABLE = 1, DISABLE = 0 for all register Enable/Disable bits

Register Table				
Address (HEX)	Name	R/W	Function	Default (HEX)
Camera Interface				
10	H_SIZE	RW	Image Width = H_SIZE x 16	14
11	V_SIZE	RW	Image Height = V_SIZE x 8	1E
12	X_OFFSETL	RW	Windows top-left X coordinate (Low) Bit[7:0]	00
13	X_OFFSETH[1:0]	RW	Windows top-left X coordinate (High) Bit[9:8]	00
14	Y_OFFSETL	RW	Windows top-left Y coordinate (Low) Bit[7:0]	00
15	Y_OFFSETH[1:0]	RW	Windows top-left Y coordinate (High) Bit[9:8]	00
16	DIVIDER	RW	Bit [7]: Speed Adjust Enable Bit [6:4]: <u>Vertical Divider</u> 000: Divide by 1 001: Divide by 2 010: Divide by 4 011: Divide by 8 100: Divide by 16 Bit [3]: LP filter Enable Bit [2:0]: <u>Horizontal Divider</u> 000: Divide by 1 001: Divide by 2 010: Divide by 4 011: Divide by 8 100: Divide by 16	00

Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
Camera Interface (Continued)				
20	DFR[6:0]	RW	<u>Data Formatter</u> Bit [6]: <u>CCIR656/CCIR601 Select</u> 0: CCIR601 1: CCIR656 Bit [5]: Reserved Bit [4]: <u>8-Bit/16-Bit Select</u>: 0: 16-Bit 1: 8-Bit Bit [3:0]: <u>Data Word Swap Formatter A Enable</u> The data output of each pixel is organized as two Words, each containing 2 bytes. The position of these 4 bytes within the 2 data words can be changed by using the Data Word Swap Formatters A (DFR[3:0]) and B (SR[7:6]). Refer to the following SR register for a description of the SR[7:6] function). MSB LSB Y0 U0 MSB LSB Y1 V0 WORD 1 WORD 2 Bit[3]: Swap U0 ↔ V0 Bit[2]: Swap Y0 ↔ Y1 Bit[1]: Swap Y1 ↔ V0 Bit[0]: Swap Y0 ↔ U0	00

Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
Camera Interface (Continued)				
21	SR	RW	<u>Synchronization Register</u> Bit [7:6]: <u>Data Word Swap Formatter B Enable</u> Bit[7]: Swap Y0 ↔ V0 Bit[6]: Swap U0 ↔ Y1 Bit [5]: <u>Buffer Synchronization 8-Bit/16-Bit Select:</u> 0: 16-Bit 1: 8-Bit Bit [4]: Buffer Synchronization Enable Bit [3]: <u>Pixel Clock Rising/Falling Edge Select</u> 0: Falling Edge 1: Rising Edge Bit [2]: Pixel Clock Delay Enable Bit [1]: HREF Delay Enable Bit [0]: Data Delay Enable	08
22	FRAR	RW	<u>Frame Rate Adjustment Register</u> Bit [7]: Defect Compensation Enable Bit [6:5]: <u>Image Mode</u> 00: YUV 01: YUV 10: RGB1 11: RGB2 Bit [4]: <u>Even Field Drop</u> 0: Drop 1: Keep Bit [3]: <u>Odd Field Drop</u> 0: Drop 1: Keep Bit [2:0]: Maximum Frame Counter Number	98

Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
Camera Interface (Continued)				
25	Format[2:0]	RW	<u>Image Format</u> Bit [2]: Bypass Bit [1:0]: <u>Format</u> 11: YUV422 10: YUV420 01: YUV411 00: YUV400	03
System Controller				
50	RESET0	RW	<u>Reset Control Register 0</u> [Reset=1] Bit [7]: GPIO Reset Bit [6]: Snapshot Reset Bit [5]: Register Reset Bit [4]: SCCB Reset Bit [3]: Audio Reset Bit [2]: Microcontroller Reset Bit [1]: UDCIF Reset Bit [0]: UDC Reset	00
51	RESET1[3:0]	RW	<u>Reset Control Register 1</u> [Reset=1] Bit [3]: Video FIFO Reset Bit [2]: JPEG Reset Bit [1]: SFIFO Reset Bit [0]: CIF Reset	00
53	EN_CLK0	RW	<u>Clock Enable 0</u> Bit [7]: Transceiver Enable Bit [6]: ISP Clock Enable Bit [5]: Reserved Bit [4]: SCCB Bit [3]: Audio Enable Bit [2]: Microcontroller Enable Bit [1]: UDCIF Enable Bit [0]: UDC Enable	87
54	En_CLK1[3:0]	RW	<u>Clock Enable 1</u> Bit [3]: Video FIFO Enable Bit [2]: JPEG Enable Bit [1]: SFIFO Enable	00

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
			Bit [0]: CIF Enable	

Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
System Controller (Continued)				
55	AUDIO_CLK[5:0]	RW	<u>Audio Clock Control</u> Bit [5]: 24MHz Clock Enable Bit [4]: Fixed phase Clock Enable Bit [3]: FIR_CLK Divide by 2 Enable Bit [2]: SD_CLK Divide by 2 Enable Bit [1:0]: <u>Clock select</u> 0: 2.048MHz 1: 2.048MHz 2: 4.096MHz 3: 6.144MHz	01
57	SNAPSHOT[5:0]	RW	<u>Snapshot Control</u> Bit [5]: Snapshot Status from Pin Bit [4]: Snapshot Status after De-Bounce Bit [3]: Snapshot Wakeup Enable Bit [2]: Host Snapshot Bit [1]: Snapshot Clear [CLEAR=1] Bit [0]: Snapshot Enable	01
58	PONOFF[4:0]	RW	<u>Power On/Off Control</u> Bit [4]: Power-On/Off Status from Pin Bit [3]: Power-On/Off Status after De-Bounce Bit [2]: Power-On/Off Wakeup Enable Bit [1]: Power-Off Status Clear [CLEAR=1] Bit [0]: Power-On/Off Enable	00
59	CAMERA_CLOCK[4:0]	RW	<u>Camera Clock Divider</u> CCLK = 48MHz / CAMERA_CLOCK [4:0] Note: 00000 = 48MHz	02
5A	YS_CTRL[6:0]	RW	<u>System Control</u> Bit [6]: Power-Down SUSPEND Mode Enable Bit [5]: USB reset Enable Bit [4]: Wakeup Enable Bit [3]: Oscillator Power-Down SUSPEND Enable -- USB Bit [2]: Oscillator Power-Down SUSPEND Enable -- μ controller Bit [1]: System Reset Mask Enable Bit [0]: System 1 1011 00 Enable	6C

Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
System Controller (Continued)				
5B	DEB_CLOCK[3:0]	RW	<u>Debounce Clock Divider</u> Bit [3:2]: <u>Debounce Clock Divider</u> 00: 0.375K 01: 0.75K 10: 1.5K 11: 96K Bit [1:0]: <u>Switching Power Clock Divider</u> 00: 24K 01: 48K 10: 96K 11: 192K	0E
5C	SYS_CLOCK[2:0]	RW	<u>System Clock Control</u> Bit [2:0]: <u>Clock Frequency</u> 000: 24MHz 001: 12MHz 010: 8MHz 011: 6MHz	00
5D	PWDN[2:0]	RW	<u>Power-Down Control</u> Bit [2]: <u>Power-Down Reset Mask</u> [Mask=1, Unmask=0] Bit [1]: Power-Down SUSPEND Mode Enable Bit [0]: <u>Operating Mode</u> 1: Normal 0: SUSPEND	02
5E	USR_DFN	RW	<u>User defined</u> <i>Users can program this register to transfer a parameter between the OV519 and PC host.</i> <u>ROM Programming:</u> Bit [7]: Program Polling Disable Bit [6:0]: Program Timeout Counter	00

Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
System Controller (Continued)				
5F	SYS_CTRL2[4:0]	R	<u>Analog Cell Control</u> Bit [4]: CEN Global Control Bit [3]: Test SUSPEND Enable Bit [2:0]: Microcontroller Reset Time Select	11
60	INTERRUPT_0[3:0]	RW	<u>Interrupt 0 (Write '1' to Clear)</u> Bit [3]: Pipe Interrupt -- USB Interrupt Bit [2]: Pipe Interrupt -- USB Bulk Out Bit [1]: Pipe Interrupt -- USB Bulk In Bit [0]: Pipe Interrupt -- USB Control	00
61	INTERRUPT_1[6:0]	RW	<u>Interrupt 1 (Write '1' to Clear)</u> Bit [6]: GPIO Bit [5]: SUSPEND Mode Bit [4]: Snapshot Bit [3]: AVG Get Bit [2]: JPEG_EOF Bit [1]: SFIFO_EOF Bit [0]: VSYNC	00
62	Mask_0[3:0]	RW	<u>Mask Control 0 (Write '1' to Clear)</u> Bit [3]: Pipe Interrupt -- USB Interrupt Bit [2]: Pipe Interrupt -- USB Bulk Out Bit [1]: Pipe Interrupt -- USB Bulk In Bit [0]: Pipe Interrupt -- USB Control	00
63	MASK_1[6:0]	RW	<u>Mask Control 1 (Write '1' to Clear)</u> Bit [6]: GPIO Bit [5]: SUSPEND Mode Bit [4]: Snapshot Bit [3]: AVG Get Bit [2]: JPEG_EOF Bit [1]: SFIFO_EOF Bit [0]: VSYNC	00
64	VCI_R0	RW	<u>Video control interrupt -- Endpoint Buffer 0</u>	00
65	VCI_R1	RW	<u>Video control interrupt -- Endpoint Buffer 1</u>	00

Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
System Controller (Continued)				
68	ADC_CTRL[3:0]	RW	<u>ADC Mute & Volume Control</u> Bit [3]: Mute Enable Bit [2:0]: AGC Gain	05
6D	UC_CTRL[1:0]	W	<u>Microcontroller Control Register</u> Bit [1]: Microcontroller SUSPEND Enable Bit [0]: Microcontroller Timer SUSPEND Enable	00
70	GPIO_LDATA_IN0	RW	GPIO Input Data0 (Level-triggered data entry)	00
71	GPIO_DATA_OUT0	RW	GPIO Output Data0	00
72	GPIO_IO_CTRL0	RW	<u>GPIO Input/Output Control0</u> 1: Input 0: Output	FF
73	GPIO_PDATA_IN0	RW	GPIO Pulse Input0 (Edge-triggered data entry)	00
74	GPIO_POLARITY0	RW	<u>GPIO Pulse Edge Polarity Select0</u> 1: Negative 0: Positive	FF
75	GPIO_PULSE_EN0	RW	<u>GPIO Pulse Enable0</u> 1: Edge-triggered data entry 0: Level-triggered data entry	00
76	GPIO_WAKEUP_EN0	RW	GPIO USB Wakeup Enable0	00
77	GPIO_RESET_MASK0	RW	GPIO Reset Mask0	00
78	GPIO_LDATA_IN1	RW	GPIO Input Data1 (Level-triggered data entry)	00
79	GPIO_DATA_OUT1	RW	GPIO Output Data1	00
7A	GPIO_IO_CTRL1	RW	<u>GPIO Input/Output Control1</u> 1: Input 0: Output	FF
7B	GPIO_PDATA_IN1	RW	GPIO Pulse Input1 (Edge-triggered data entry)	00
7C	GPIO_POLARITY1	RW	<u>GPIO Pulse Edge Polarity Select1</u> 1: Negative 0: Positive	FF
7D	GPIO_PULSE_EN1	RW	<u>GPIO Pulse Enable1</u> 1: Edge-triggered data entry 0: Level-triggered data entry	00
7E	GPIO_WAKEUP_EN1	RW	GPIO USB Wakeup Enable1	00
7F	GPIO_RESET_MASK1	RW	GPIO Reset Mask1	00

Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
System Controller (Continued)				
80	GPIO_LDATA_IN2	RW	GPIO Input Data2 (Level-triggered data entry)	00
81	GPIO_DATA_OUT2	RW	GPIO Output Data2	00
82	GPIO_IO_CTRL2	RW	<u>GPIO Input/Output Control2</u> 1: Input 0: Output	FF
83	GPIO_PDATA_IN2	RW	GPIO Pulse Input2 (Edge-triggered data entry)	00
84	GPIO_POLARITY2	RW	<u>GPIO Pulse Edge Polarity Select2</u> 1: Negative 0: Positive	FF
85	GPIO_PULSE_EN2	RW	<u>GPIO Pulse Enable2</u> 1: Edge-triggered data entry 0: Level-triggered data entry	00
86	GPIO_WAKEUP_EN2	RW	GPIO USB Wakeup Enable2	00
87	GPIO_RESET_MASK2	RW	GPIO Reset Mask2	00
88	GPIO_IRQ_EN0	RW	GPIO Interrupt Enable0	00
89	GPIO_IRQ_EN1	RW	GPIO Interrupt Enable1	00
8A	GPIO_IRQ_EN2	RW	GPIO Interrupt Enable2	00
8B	GPIO_IRQ_EN3	RW	GPIO Interrupt Enable3	00
8C	IO_N[4:0]	RW	<u>I/O Pad In/Out Control</u> [Inputs=1, Outputs=0] Bit [4]: UV/GPIO Port Bit [3]: Sensor Bit [2]: Microcontroller Data Port Bit [1]: Microcontroller P2 Bit [0]: Microcontroller Control Port (/PSRD, /PSDWR, /PSCE, /CE, /OE, /WE)	1F
8D	IO_Y[4:0]	RW	<u>I/O Output Data</u> Bit [4]: UV/GPIO port Bit [3]: Sensor Bit [2]: Microcontroller data port Bit [1]: Microcontroller P2 Bit [0]: Microcontroller control port (/PSRD, /PSDWR, /PSCE, /CE, /OE, /WE)	00
Y Enhancement				
A8	OFFSET	RW	Offset	00
A9	GAIN	RW	Gain	00
AA	BRIGHTNESS	RW	Brightness	00
Average				
B0	AVG_CTRL[1:0]	R/W	<u>Image Processing Control</u>	00

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
			Bit [1]: Y Distribution Enable Bit [0]: Averaging Enable	
B1	AVG_HSA[5:0]	R/W	<u>H Start Address for Window</u> H Start Address = AVG_HSA[5:0] / 16	0A
B2	AVG_VSA[6:0]	R/W	<u>V Start Address for Window</u> Real V Start Address = AVG_VSA[6:0] / 8	0F
B3	AVG_HEA[5:0]	R/W	<u>H End Address for Window</u> Real V Start Address = AVG_HEA[5:0] / 18	1E
B4	AVG_VEA[6:0]	R/W	<u>V End Address for Window</u> Real V Start Address = AVG_VEA[6:0] / 8	3D

Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
Average (Continued)				
B5	AVG_YREFH	R/W	Y Data High Reference	FF
B6	AVG_YREFL	R/W	Y Data Low Reference	00
B7	AVG_UREFH	R/W	U Data High Reference	FF
B8	AVG_UREFL	R/W	U Data Low Reference	00
B9	AVG_VREFH	R/W	V Data High Reference	FF
BA	AVG_VREFL	R/W	V Data Low Reference	00
BB	AVG_Y	R	Y Average	
BC	AVG_U	R	U Average	
BD	AVG_V	R	V Average	
Defect Compensation				
C0	H0H[1:0]	RW	Defect 0 H address, H0[9:8]	03
C1	H0L	RW	Defect 0 H address, H0[7:0]	FF
C2	V0H[1:0]	RW	Defect 0 V address, V0[9:8]	03
C3	V0L	RW	Defect 0 V address, V0[7:0]	FF
C4	H1H[1:0]	RW	Defect 1 H address, H1[9:8]	03
C5	H1L	RW	Defect 1 H address, H1[7:0]	FF
C6	V1H[1:0]	RW	Defect 1 V address, V1[9:8]	03
C7	V1L	RW	Defect 1 V address, V1[7:0]	FF
C8	H2H[1:0]	RW	Defect 2 H address, H2[9:8]	03
C9	H2L	RW	Defect 2 H address, H2[7:0]	FF
CA	V2H[1:0]	RW	Defect 2 V address, V2[9:8]	03
CB	V2L	RW	Defect 2 V address, V2[7:0]	FF
CC	H3H[1:0]	RW	Defect 3 H address, H3[9:8]	03
CD	H3L	RW	Defect 3 H address, H3[7:0]	FF
CE	V3H[1:0]	RW	Defect 3 V address, V3[9:8]	03
CF	V3L	RW	Defect 3 V address, V3[7:0]	FF
D0	H4H[1:0]	RW	Defect 4 H address, H4[9:8]	03
D1	H4L	RW	Defect 4 H address, H4[7:0]	FF
D2	V4H[1:0]	RW	Defect 4 V address, V4[9:8]	03
D3	V4L	RW	Defect 4 V address, V4[7:0]	FF
D4	H5H[1:0]	RW	Defect 5 H address, H5[9:8]	03
D5	H5L	RW	Defect 5 H address, H5[7:0]	FF
D6	V5H[1:0]	RW	Defect 5 V address, V5[9:8]	03
D7	V5L	RW	Defect 5 V address, V5[7:0]	FF

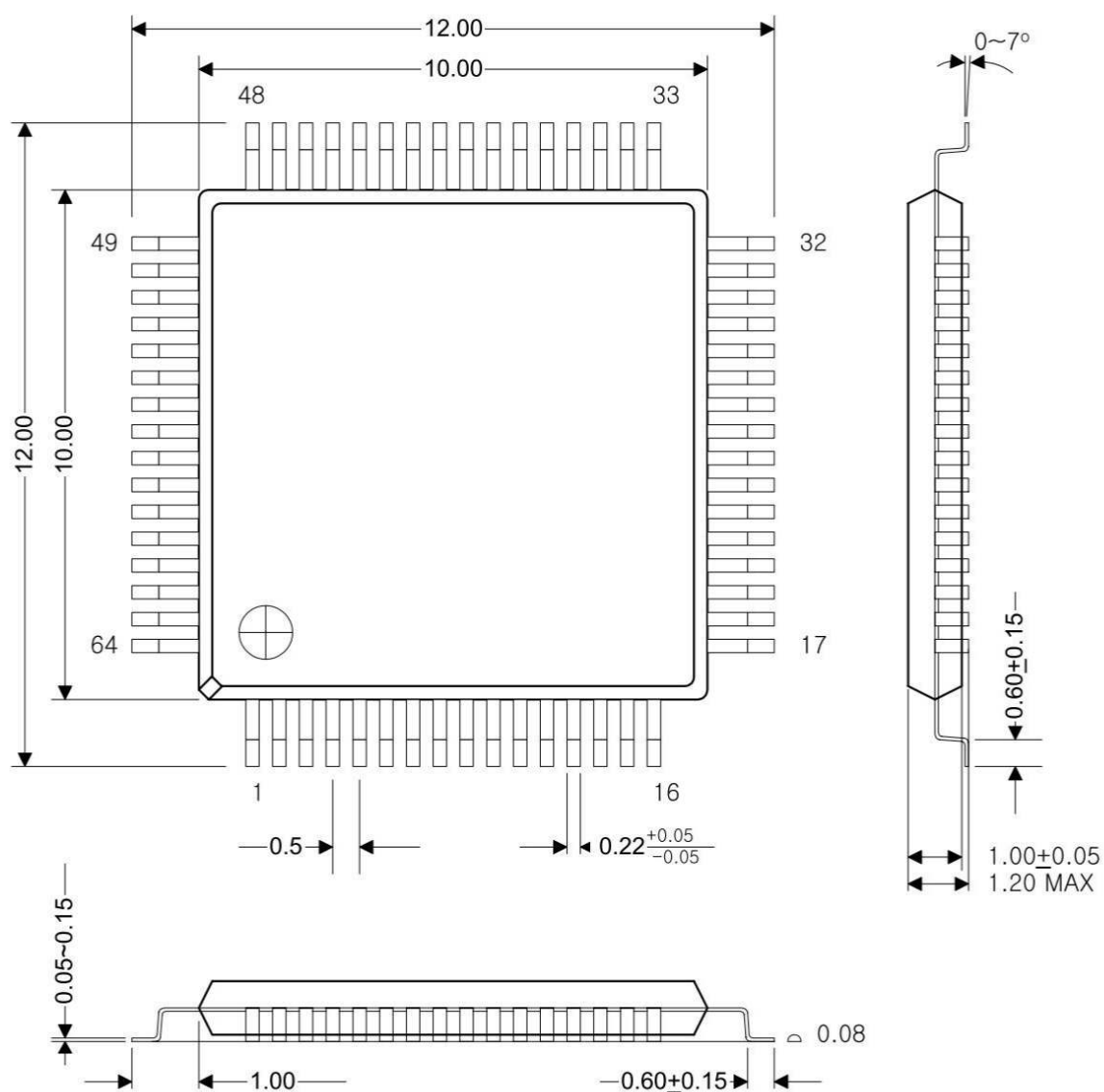
Device Control Registers (Continued)

Register Table (Continued)				
Address (HEX)	Name	R/W	Function	Default (HEX)
Defect Compensation (Continued)				
D8	H6H[1:0]	RW	Defect 6 H address, H6[9:8]	03
D9	H6L	RW	Defect 6 H address, H6[7:0]	FF
DA	V6H[1:0]	RW	Defect 6 V address, V6[9:8]	03
DB	V6L	RW	Defect 6 V address, V6[7:0]	FF
DC	H7H[1:0]	RW	Defect 7 H address, H7[9:8]	03
DD	H7L	RW	Defect 7 H address, H7[7:0]	FF
DE	V7H[1:0]	RW	Defect 7 V address, V7[9:8]	03
DF	V7L	RW	Defect 7 V address, V7[7:0]	FF
Y Distribution (User Histogram Data)				
F0	REF0	RW	Y Distribution: Reference 0	20
F1	REF1	RW	Y Distribution: Reference 1	40
F2	REF2	RW	Y Distribution: Reference 2	60
F3	REF3	RW	Y Distribution: Reference 3	80
F4	REF4	RW	Y Distribution: Reference 4	A0
F5	REF5	RW	Y Distribution: Reference 5	C0
F6	REF6	RW	Y Distribution: Reference 6	E0
F7	YD0	R	Number of Y Data: Region = 0 ↔ REF0	00
F8	YD1	R	Number of Y Data: Region = REF0 ↔ REF1	00
F9	YD2	R	Number of Y Data: Region = REF1 ↔ REF2	00
FA	YD3	R	Number of Y Data: Region = REF2 ↔ REF3	00
FB	YD4	R	Number of Y Data: Region = REF3 ↔ REF4	00
FC	YD5	R	Number of Y Data: Region = REF4 ↔ REF5	00
FD	YD6	R	Number of Y Data: Region = REF5 ↔ REF6	00
FE	YD7	R	Number of Y Data: Region = REF6 ↔ 0xFF	00

Factory-Reserved Registers (Do Not Access)	
00 → 0F	69 → 6C
17 → 1F	6E → 6F
23 → 24	8E → A7
26 → 4F	AB → AF
52	BE → BF
56	E0 → EF
66 → 67	FF

OV519 TQFP-64 Package:

Dimension: mm



Note:

- All information shown herein is current as of the revision and publication date. Please refer to the OmniVision web site (<http://www.ovt.com>) to obtain the current versions of all documentation.
- OmniVision Technologies, Inc. reserves the right to make changes to their products or to discontinue any product or service without further notice (It is advisable to obtain current product documentation prior to placing orders).
- Reproduction of information in OmniVision product documentation and specifications is permissible only if reproduction is without alteration and is accompanied by all associated warranties, conditions, limitations and notices. In such cases, OmniVision is not responsible or liable for any information reproduced.
- This document is provided with no warranties whatsoever, including any warranty of merchantability, non-infringement, fitness for any particular purpose, or any warranty otherwise arising out of any proposal, specification or sample. Furthermore, OmniVision Technologies Inc. disclaims all liability, including liability for infringement of any proprietary rights, relating to use of information in this document. No license, expressed or implied, by estoppels or otherwise, to any intellectual property rights is granted herein.
- 'OmniVision', 'CameraChip' are trademarks of OmniVision Technologies, Inc. All other trade, product or service names referenced in this release may be trademarks or registered trademarks of their respective holders. Third-party brands, names, and trademarks are the property of their respective owners.

For further information, please feel free to contact OmniVision at info@ovt.com.

**OmniVision Technologies, Inc
Sunnyvale, CA USA
(408) 733-3030**

